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github.com/aryashreep/learn-devops-in-90-days

DevOps & Cloud Training Series · 2026
Edition



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LINUX FOUNDATIONS

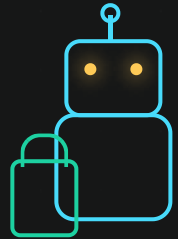
Day 02: Under the Hood of DevOps Power OS 🐧



Golu

"Ladoo, sab kehte hain ki DevOps matlab Linux hai. Par ye Linux actually hai kya? Bas ek kala (black) screen?"

"Haha! Nahi Golu, Linux sirf screen nahi, pura **Operating System ka Engine** hai! Aaj hum iska system breakdown karenge."



Ladoo

Bhai, Aaj Kya Seekhenge?

Aaj hum Linux ke **Core Concepts** samjhenge: Kernel, Shell, aur OS layers. Hum ye bhi janenge ki kyu 90% cloud Linux par chalta hai aur distributions (distros) kya hote hain!



LEARNING OBJECTIVES

- **Define:** What is an OS and what are its responsibilities.
- **Distinguish:** Kernel vs Shell (The two core pillars).
- **Navigate:** Understand the Onion Model of system layers.
- **Identify:** Common Linux Distros like Ubuntu, CentOS, and Alpine.

💡 **Kyu Seekhein?** Kyunki cloud computing, containers (Docker), aur modern servers bina Linux ke survive nahi kar sakte!

💡 ANALOGY: THE RESTAURANT MANAGER



Golu

"Computer ek **Restaurant** jaisa hai! Hardware Kitchen hai, Apps Customers hain."

1. **Kitchen (Hardware)**: CPU, RAM, Disk.
2. **Customers (Applications)**: Browsers, Spotify, DBs.
3. **Manager (OS)**: Jo customer se order leta hai aur kitchen se kaam karwata hai (without chaos!).

? DID YOU KNOW?

90%+

Cloud Servers run Linux

33yr

Linux since 1991

600+

Linux Distros exist

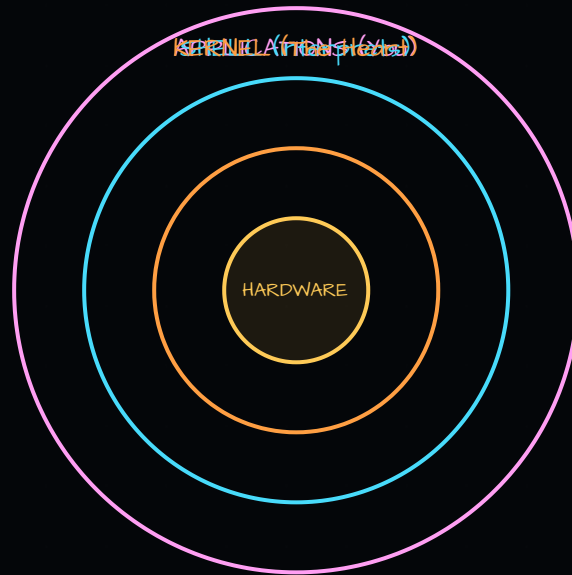


KEY DIRECTORIES — WHAT LIVES WHERE

-  **/ (Root)** → The trunk of the entire system tree.
 -  **/bin** → Essential user command binaries (ls, cp).
 -  **/etc** → System configuration files.
 -  **/var/log** → Variable data like system logs.
-



KEY CONCEPTS: SYSTEM LAYERS



Bhai simple hai: Kernel hardware se baat karta hai, Shell aapse baat karti hai!



COMPARE: WITH VS WITHOUT OS

Without OS (Chaos)

"Har app khud RAM aur CPU control karti. Ek crash hua toh pura system dead!"

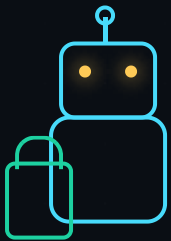


With Linux OS

"Central management, isolated processes, aur automatic crash recovery!"



REAL-WORLD EXAMPLE: NETFLIX



"Netflix ke paas **Zero Windows Servers** hain production mein! Wo 100% Linux use karte hain scaling ke liye."

- **Situation:** Needed to serve 260M+ users globally.
- **Solution:** Used Linux Kernel features (cgroups, namespaces) for massive scaling.

⚠ COMMON MISCONCEPTIONS

- **Myth:** "Linux" Windows jaisa ek Operating System hai.
- **Reality:** Linux actually ek **Kernel** hai. Ubuntu/CentOS Distros hain!
- **Myth:** GUI-less (CLI) Linux incomplete hota hai.
- **Reality:** CLI servers fast, secure, aur reliable hote hain. Ye Professional standard hai!

▶ TRY IT YOURSELF

1. `uname -r` *(Kernel version dekho)*
2. `cat /etc/os-release` *(Tumhara Distro kaunsa hai?)*
3. `whoami && hostname` *(Tum kaun ho aur machine ka naam kya hai?)*

✓ QUICK SUMMARY

- **Kernel:** Software heart (Talks to hardware).
 - **Shell:** User interface (Interprets your commands).
 - **Distro:** Kernel + Bundled tools (Ubuntu, Debian, etc.).
 - **Linux Power:** Open-source, Secure, and Cloud-native.
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? MINI QUIZ — TEST YOUR UNDERSTANDING

1. Kernel kisse direct baat karta hai?

- a) Apps
- b) Hardware
- c) User
- d) Screen

2. AWS EC2 free tier mein standard distro kaunsa hai?

- a) Alpine
- b) Debian
- c) Ubuntu
- d) Red Hat

3. Sabse chota (5MB) Linux distro kaunsa hai?

- a) CentOS
- b) Alpine
- c) Ubuntu
- d) Windows

4. Kernel version check karne ka command?

- a) `uname -r`
- b) `whoami`
- c) `cat log`
- d) `ls -la`

5. Shell ka kaam kya hai?

- a) Edit files
- b) Interpret commands
- c) Manage CPU
- d) Storage

6. Onion model ke sabse outer layer mein kya hota hai?

- a) Kernel
- b) Hardware
- c) Applications
- d) Shell

7. `/etc` directory mein kya milti hai?

- a) Commands
- b) Logs
- c) Personal files
- d) Config files

✓ ANSWER KEY

"Golu, match kar lo: 1-b | 2-c | 3-b | 4-a | 5-b | 6-c | 7-d"

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